

Geometry

Unit 12

Lesson 6 Homework

Name _____

Date _____

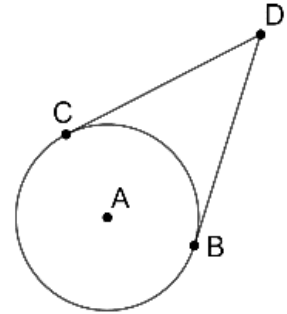
Period _____

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 3x + 8$ and $DB = 5x - 16$.

1. What is the value of x ?

2. What is the length of $\overline{CD}$?

3. What is the length of $\overline{DB}$?



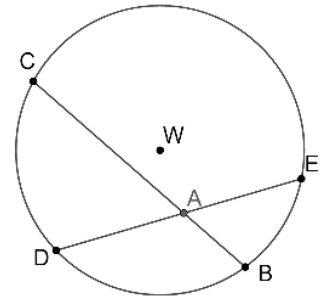
For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A . $CA = 3x + 5$, $AB = 3$, $DA = 4$, and $AE = 2x + 4$.

4. Write an equation you can use to solve for x .

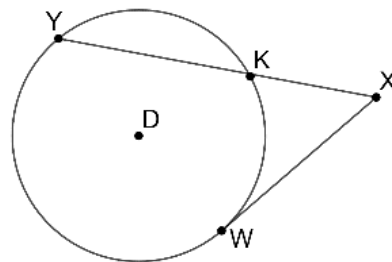
5. Solve the equation you wrote in question 4 for x .

6. What is the length of $\overline{CB}$?

7. What is the length of $\overline{DE}$?



For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle, and intersects tangent $\overline{XW}$ at point X. The length of $\overline{XW} = 6$, $\overline{KX} = 2$, and $\overline{YX} = 3x + 1$.



8. Write an equation that can be used to find the value of x .

9. Solve the equation you wrote in question 8 to find the value of x .

10. What is the length of $\overline{YX}$?